High-Quality Angle Prediction for Oriented Object Detection in Remote Sensing Images

Guanchun Wang, Xiangrong Zhang, *Senior Member, IEEE*, Peng Zhu, Xu Tang, *Senior Member, IEEE*, Puhua Chen, *Senior Member, IEEE*, Licheng Jiao, *Fellow, IEEE*, and Huiyu Zhou

***Abstract*— Oriented object detection is a challenging task in remote sensing, where the detected objects can be represented by oriented bounding boxes (OBBs). Angle prediction in oriented object detection has been widely studied, due to its crucial role in object detection. However, the precision of angle prediction is severely limited by misalignments in most of the existing methods, including representation-, evaluation-, and optimization-based misalignments. To alleviate these misalignments, this article presents a novel angle prediction method, called angle quality estimation (AQE). Specifically, our proposed AQE transforms the angle prediction task into a distribution estimation task to address the representation misalignment problem and implicitly measure the quality of the predicted angles. Based on the estimated AQ, we then propose a new metric to comprehensively evaluate the quality of OBBs. Then, we propose an object aspect ratio-based loss function to optimize angle prediction for addressing the optimization misalignment. Our proposed AQE is a plug-and-play method, which can be embedded on any existing oriented object detector. Experimental results on three public benchmarks, including dataset for object detection in aerial images (DOTA), high resolution ship collections 2016 (HRSC2016), and International Conference on Document Analysis and Recognition 2015 (ICDAR2015) datasets, show that our method achieves better performance than the other state of the arts.**

***Index Terms*— Angle quality estimation (AQE), arbitrary-oriented object detection, remote sensing images (RSIs).**

I. Introduction

With the rapid development of computer vision, object detection has attracted widespread attention [1], [2], [3], [4], [5], [6], [7]. Along with the advances in convolutional neural network (CNN), a series of general object detection methods, such as [8], [9], [10], and [11], have been established and achieved great progress. However, different from the general object detection, objects in remote sensing images (RSIs) are usually arbitrary-oriented; thereby, horizontal bounding boxes (HBBs) are incapable to precisely represent them, resulting in background interference.

To address this issue, many oriented object detection methods [12], [13], [14], [15], [16], [17] that employ oriented bounding boxes (OBBs) to represent objects are proposed and achieved promising progress. Angle prediction plays a crucial role in oriented object detection, as it is the essential difference between HBBs and OBBs. However, there are still three misalignments in the most existing methods.

1) Representation Misalignment: The existing methods usually represent the angle as a regular variable without considering its periodicity and directly minimize the $L1$ distance between the predicted angle and the truth during the training process. This leads to a misalignment between angle periodicity and the monotonicity of the objective function. In particular, as shown in the first row of Fig. 1, although the prediction (blue OBB) is quite close to the ground truth (green OBB), there is still a large $L1$ distance between them. In addition, the red and blue OBBs have the same intersection over union (IoU) with the ground truth but exhibit a large discrepancy on $L1$ distance.

A. Evaluation Misalignment

In most previous studies, only the classification scores are employed as the criterion for evaluating detection results; however, high classification scores cannot guarantee accurate localization results, called evaluation misalignment, as illustrated in Fig. 1. In contrast, we observe that the quality of the predicted angle is a crucial criterion for evaluating the localization results of detections while is neglected by the existing methods.

B. Optimization Misalignment

Some previous works underestimate that the role of angle prediction is greatly associated with the aspect ratio of objects. As shown in the bottom right of Fig. 1, the variation of angle plays a significant role in localization precision for objects with high aspect ratios and has almost no influence on objects with low aspect ratios, which is statistically demonstrated in Section III-C. However, objects with different aspect ratios are optimized with the same intensity of angle supervision, resulting in a optimization misalignment.

To deal with these problems, we here propose a novel angle prediction approach, called angle quality estimation (AQE). Specifically, our AQE transforms the angle prediction task into a distribution estimation task via a simple but ingenious modeling process, which implicitly estimates the quality of the predicted angle. Through modeling the angle as a periodic distribution, the representation misalignment is avoided, which significantly improves the precision of angle prediction. Based on the estimated AQ, we propose an AQ-driven non-maximum suppression (AQ-NMS) method to rescore the detection results

Manuscript received 11 January 2023; revised 18 May 2023; accepted 28 June 2023. Date of publication 4 July 2023; date of current version 19 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62276197 and Grant 62171332 and in part by the Royal Society-Newton Advanced Fellowship under Grant NA160342. *(Corresponding author: Xiangrong Zhang.)*

Guanchun Wang, Xiangrong Zhang, Peng Zhu, Xu Tang, Puhua Chen, and Licheng Jiao are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: xrzhang@mail.xidian.edu.cn).

Huiyu Zhou is with the School of Computing and Mathematical Sciences, University of Leicester, LE1 7RH Leicester, U.K.

Digital Object Identifier 10.1109/TGRS.2023.3292111

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

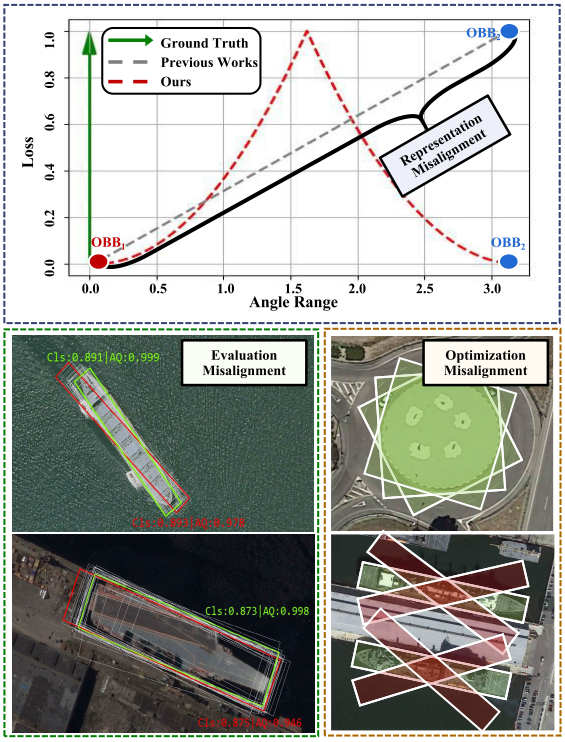


Fig. 1. Visualization of three limitations of the existing methods for oriented object detection: 1) the representation misalignment is shown in the first row, where the red and blue OBBs have the same IoU with the ground truth but a large difference in loss due to angle periodicity; 2) as shown in the bottom left, high classification scores (Cls) cannot guarantee the accurate localization result, where AQ indicates the angle quality estimated by the proposed method; and 3) as shown in the bottom right, angle prediction plays a more crucial role in localization precision for objects with high aspect ratios than those with low aspect ratios.

by combining the AQ and the classification score, thereby rectifying the evaluation misalignment.

Furthermore, we quantitatively analyze the relationship between the angle sensitivity and the aspect ratio of objects. On the basis of this, we design a novel angle loss function, named aspect-ratio-aware angle loss (Triple-A Loss), to mitigate the optimization misalignment by dynamically optimizing the angle prediction task, which further enhances the detection performance. Finally, we evaluate the proposed method on three public arbitrary-oriented detection datasets (i.e., dataset for object detection in aerial images (DOTA) [18], HRSC2016 [19], and International Conference on Document Analysis and Recognition 2015 (ICDAR2015) [20]). Compared with the existing angle prediction approach, our method achieves better performance.

The contributions can be summarized as follows.

1) A novel angle prediction approach named AQE is proposed. By modeling the angle as a periodicity distribution, the angle prediction task is converted into a distribution estimation task, avoiding the representation misalignment and estimating the quality of the predicted angle.
2) Based on the estimated AQ, an AQ-NMS is designed to rescore the detection results, rectifying the misalignment between classification scores and localization precision.
3) A Triple-A Loss function is proposed to adaptively adjust the loss weights according to the aspect ratio of objects. In this way, the optimization misalignment is eliminated, which further enhances the model performance.

The remainder of this article is organized as follows. In Section II, we briefly review related works on horizontal object detection, oriented object detection, and regression quality estimation methods. In Section III, we describe the details of our proposed detection framework AQE-Det. We introduce the dataset and evaluation metrics in Section IV and show the experiment settings and results in Section V. Finally, Section VI concludes this article.

II. Related Work

A. Horizontal Object Detection

Horizontal object detection methods [9], [10], [11], [21], [22], [23], [24], [25] have achieved great progress in recent years, where HBBs are utilized to represent objects. Most of them can be categorized into two groups: two-stage and single-stage methods. The two-stage methods first generate a series of candidate proposals for potential objects and then feed the proposals to the detection head for refining the locations and categories of objects. The classical two-stage method Faster-region-based CNN (RCNN) [9] is widely used as a basis in the future research [10], [23] due to its effectiveness, in which a region proposal network (RPN) is utilized to generate candidate proposals and a region-of-interest (RoI) pooling layer [8] is adopted to extract proposal features. Ren et al. [9] introduced an auxiliary segmentation branch to improve the detection performance, and Cai et al. proposed to refine the detection results with a cascade framework. To satisfy real-time demand in practice, many one-stage methods [21], [22] have been proposed. Redmon et al. [21] follow a regression way that directly predicts the category and location of objects in each image, without any region proposal. Taking the inspiration from RPN and multiscale technique, Liu et al. [22] proposed a simple but effective detector single shot detector (SSD), achieving faster speed and better accuracy. Benefiting from the powerful feature representation ability of CNN, these methods achieve successful progress in general object detection. However, the objects in RSIs are often densely arranged and arbitrary-oriented; thereby, horizontal bounding boxes are incapable of precisely locating them.

B. Oriented Object Detection

To overcome the limitation of horizontal object detection methods, a series of oriented object detection methods have been proposed [12], [13], [14], [15], [16], [17], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]. Jiang et al. [12] proposed a rotational region CNN (R2CNN) method, which performs oriented object detection by adding an angle prediction branch into [9]. However, HBBs are still utilized in the RPN stage, resulting in imprecise of object feature extraction. To address this problem, Ma et al. [13] extend the vanilla RPN to a rotation framework by introducing rotated anchors, enhancing the quality of the proposal generation. Similar to [13], Yang et al. [14] proposed the rotation dense feature pyramid networks (R-DFPNs), where a rotation region detection network and a densely connected feature fusion module are designed to solve the issue of oriented detection and scale variation, respectively. Ding et al. [15] proposed an

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

RoI Transformer to learn the rotation parameters from HBBs; thereby, the network can accurately extract the features from rotated regions, which greatly improves the precision of detection. Unlike the above two-stage methods, Yang et al. [16] proposed a single-stage detector to achieve both high speed and accuracy by introducing a feature refine module and a rotation detection module. Considering the drawbacks in the existing label assignment strategies, Ming et al. [36] proposed a dynamic anchor selection strategy to mitigate the misalignment between classification score and localization precision in oriented object detection, and Hou et al. [37] proposed a object-shape-driven label assignment strategy for improving detector.

Although the above methods achieve outstanding performance, they do not consider the representation misalignment, i.e., the loss value at the angular boundary is usually mutated, resulting in the instability of localization. To deal with this issue, some methods [31], [32], [33], [34] have been proposed. Yang et al. [31] proposed an IoU-based $L1$ loss, which alleviates the representation misalignment by introducing an IoU factor. Qian et al. [32] directly regressed the four points of OBBs and proposed a modulated Loss, which avoids the representation misalignment by reordering the corner points. To address this problem from the source, Yang and Yan [33] proposed to transform angle prediction into the angle classification task by adding an angular classification head and further designed a densely coded label [34] to reduce computational cost caused by the definition range of angle is vast. In this article, we investigate the misalignments at different levels in the angle prediction task and propose a more effective approach for angle prediction, which avoids these misalignments with almost no extra cost.

C. Localization Quality Estimation

The localization quality of the detection results is essential but often forgotten in the existing object detection methods. They usually assess detection results in the NMS process according to classification scores. Nevertheless, high classification scores cannot guarantee the precise localization of detections due to the misalignment between classification scores and localization precision, resulting in some precise detections are suppressed. To deal with this problem, some methods have been proposed [38], [39], [40], [41], [42], [43]. Jiang et al. [38] proposed to predict the IoU of HBBs and utilize it to enhance the postprocessing of detection results, while the training process of the IoU prediction branch is too complex. He et al. [39] proposed to model HBBs as a Gaussian distribution and designed a novel Kullback-Leibler divergence (KL) loss for learning the variance of HBBs, which can be served as the localization quality. Similar to [39], Choi et al. [40] proposed to model HBB in you only look once (YOLO) as a Gaussian distribution to estimate the uncertainty of the detection results, alleviating the mislocalization issue in autonomous driving applications. Lee et al. [41] proposed a Gaussian fully convolutional one-stage object detector (FCOS) [41] framework, which introduces the uncertainty estimation into the anchor-free detector and proposes an uncertainty loss for learning the variance of HBBs. To further estimate the localization quality, Li et al. [43] proposed to model the parameters of HBBs as arbitrary distributions, rectifying the mislocalization issue. Since these methods exhibit effectiveness in dealing with HBBs, we take inspiration from them and attempt to solve the crucial issues of oriented object detection by using AQE.

III. Proposed Method

In this section, we first introduce our angle prediction method in detail. Then, an AQ-based NMS method is proposed to improve the localization precision of detector. Finally, we propose an aspect-ratio-aware loss for training the angle prediction branch.

A. AQ Estimation

Angle prediction is a crucial but challenging problem in oriented object detection. Most of the existing angle prediction methods follow a regression manner to predict angles, which modify the horizontal object detector by adding an extra angle parameter in the regression branch, resulting in the representation and evaluation misalignments as discussed above. To this end, we design a novel angle prediction approach, namely, AQE, which addresses the above drawbacks by modeling the angle prediction task as a probability distribution estimation task. The architecture of the AQE is illustrated in Fig. 2.

1) Modeling Phase: In modeling phase we employ a periodic Gaussian distribution to represent the angular variable for simplicity. Specifically, an AQE branch consisting of two parallel fully connected layers is proposed for predicting the mean μ and variance σ of angle as follows:

$$\mu, \sigma = \mathrm{FC}_{\mathrm{AQE}}\left(F_{\mathrm{obj}}\right) \tag{1}$$

where $\mathrm{FC}_{\mathrm{AQE}}$ is the AQE branch and F_{obj} is the features of object regions. However, the two variables are unbounded, resulting in the difficulty of optimization. We, thus, set up the definition range of angle in AQE between 0 and π and restrict the standard deviation between 0 and 1 to facilitate the evaluation of the AQ as follows:

$$\begin{aligned} \theta_\mu &= \pi \cdot \mathrm{Sigmoid}(\mu) \\ \theta_\sigma &= \mathrm{Sigmoid}(\sigma). \end{aligned} \tag{2}$$

Then, a 1-D Gaussian distribution with periodicity is constructed for the angular variable

$$P(\theta) = \mathrm{Exp}\left(-\frac{\left(\theta - \theta_\mu - \alpha \cdot \pi\right)^2}{2\theta_\sigma^2}\right), \quad \theta \in [0, \pi] \tag{3}$$

$$\alpha = \begin{cases} 0, & |\theta - \theta_\mu| < \dfrac{\pi}{2} \\ \mathrm{Sign}\left(\theta - \theta_\mu\right), & |\theta - \theta_\mu| > \dfrac{\pi}{2}, \end{cases} \quad \theta \in [0, \pi] \tag{4}$$

where θ_μ and θ_σ are the mean and standard deviation of angles, respectively, and α is the periodic factor. θ_σ reveals the quality of the predicted angle. A lower θ_σ means the predicted angle is more credible. Similarly, we model the ground truth of angle as a Dirac distribution

$$\delta(\theta) = \begin{cases} 1, & \theta = \theta_{\mathrm{gt}} \\ 0, & \text{elsewise}, \end{cases} \quad \theta \in [0, \pi]. \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

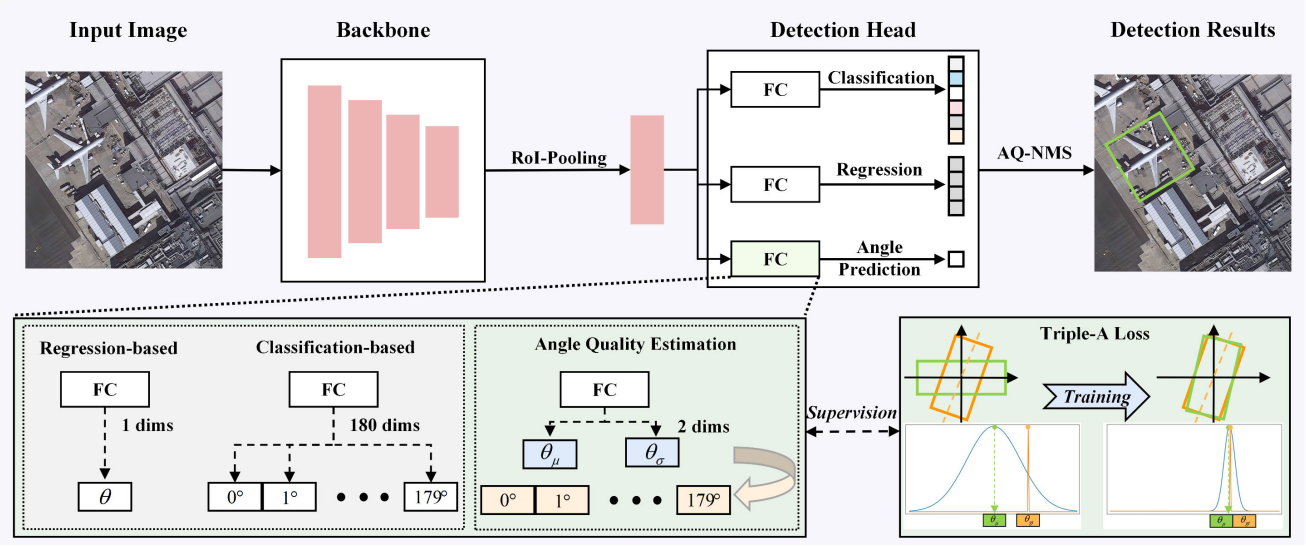


Fig. 2. Overview of the proposed method, which consists of a regular object detector and the proposed AQE method. Our AQE addresses the representation misalignment and estimate the quality of the predicted angles by representing angle as a Gaussian distribution with almost no computational costs. In comparison with the existing methods, our AQE exhibits superiority in enhancing the precision of angle prediction (compared with regression-based) and alleviating computational costs (compared with classification-based). It is worth noting that AQE is a plug-and-play method that can be easily integrated with arbitrary object detectors.

After modeling, we transform the angle regression task into a distribution approximation task. Its goal is to estimate the model parameters for approximating the predicted distribution $P(\theta)$ to the ground-truth distribution $\delta(\theta)$ with N training samples. The task can be expressed in the following formula:

$$\Theta = \arg\min_{\Theta} \frac{1}{N} \sum_{i \in N} \text{Distance}(\delta(\theta_i) \| P(\theta_i; \Theta)) \tag{6}$$

where Distance$(*\|*)$ is the measure of the gap between two distributions. In this way, the representation misalignment is avoided by representing the angle as a Gaussian distribution with periodicity, as shown in Fig. 2.

2) Training Phase: Obviously, it is challenging to directly approximate two distributions by parameterized methods. We, thus, transfer the continuous distribution into discrete counterparts for the purposes of making calculations more easier. Formally, we sample the above angle distribution with equal intervals on the continuous distribution, acquiring a Gaussian label with length M, called $L_{\text{G}}(x)$

$$L_{\text{G}}(x) = \text{Exp}\left(-\frac{\left(x - \theta_\mu - \alpha \cdot \pi\right)^2}{2\theta_\sigma^2}\right) \tag{7}$$

where $x = [0, (\pi/M), \ldots, ((M-1)\pi)/M]$ and the length of the Gaussian label is calculated as follows:

$$M = \frac{180}{R_\theta} \tag{8}$$

where R_θ is the angle resolution. Similarly, the ground truth of the angle is encoded into a one-hot vector of length M

$$L_{\text{gt}}(x) = \begin{cases} 1, & x = \theta_{\text{gt}} \\ 0, & \text{elsewise.} \end{cases} \tag{9}$$

In this way, we can simply solve the original task by serving it as a classification problem, which is easy to be optimized.

3) Inference Phase: During the inference phase, we directly take θ_μ as the predicted angles and produce detection results. On the other hand, we leverage θ_σ to generate the estimated quality of the predicted angle

$$\text{quality}_{\text{angle}} = 1 - \theta_\sigma. \tag{10}$$

Then, the detection results can be comprehensively evaluated with a combination of classification score and AQ in Section III-B.

4) Theoretical Analysis: In this part, we develop an analysis of why θ_σ can be considered as the quality of predicted angle. Considering the cross entropy loss as an example, the objective function in (6) can be transformed as follows:

$$\begin{aligned}
\Theta &= \underset{\Theta}{\text{argmin}} \frac{1}{N} \sum_{i \in N} \text{Distance}(\delta(\theta_i) \| P(\theta_i; \Theta)) \\
&\propto \underset{\Theta}{\text{argmin}} \frac{1}{N} \sum_{i \in N} \text{CrossEntropy}\left(L_{\text{G},i}, L_{\text{gt},i}\right) \\
&= \underset{\Theta}{\text{argmin}} \frac{1}{N} \sum_{i \in N} -\log \frac{\text{Exp}\left(L_{\text{G},i}\left(\theta_{\text{gt}}\right)\right)}{\sum_{x=0}^{\frac{M-1}{M}\cdot\pi} \text{Exp}\left(L_{\text{G},i}(x)\right)} \\
&= \underset{\Theta}{\text{argmin}} \frac{1}{N} \sum_{i \in N} \log\left(1 + \sum_{x=0, x \neq \theta_{\text{gt}}}^{\frac{M-1}{M}\cdot\pi} e^{\left(L_{\text{G},i}(x) - L_{\text{G},i}\left(\theta_{\text{gt}}\right)\right)}\right)
\end{aligned} \tag{11}$$

where $\sum_{x=0, x\neq\theta_{\text{gt}}}^{\frac{M-1}{M}\cdot\pi} \text{Exp}(L_{\text{G},i}(x) - L_{\text{G},i}(\theta_{\text{gt}}))$ can be simplified and decomposed as follows:

$$\begin{aligned}
&\sum_{x=0, x \neq \theta_{\text{gt}}}^{\frac{M-1}{M}\cdot\pi} \text{Exp}\left(L_{\text{G}}(x) - L_{\text{G}}\left(\theta_{\text{gt}}\right)\right) \\
&= \text{Exp}\left(e^{\frac{(0-\theta_\mu-\alpha\cdot\pi)^2}{-2\theta_\sigma^2}} - e^{\frac{(\theta_{\text{gt}}-\theta_\mu-\alpha\cdot\pi)^2}{-2\theta_\sigma^2}}\right)
\end{aligned}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

$$+\operatorname{Exp}\left(e^{\frac{\left(\frac{\pi}{M}-\theta_\mu-\alpha\cdot\pi\right)^2}{-2\theta_\sigma^2}}-e^{\frac{(\theta_{\mathrm{gt}}-\theta_\mu-\alpha\cdot\pi)^2}{-2\theta_\sigma^2}}\right)$$

$$\vdots$$

$$+\operatorname{Exp}\left(e^{\frac{\left(\frac{(M-1)\cdot\pi}{M}-\theta_\mu-\alpha\cdot\pi\right)^2}{-2\theta_\sigma^2}}-e^{\frac{(\theta_{\mathrm{gt}}-\theta_\mu-\alpha\cdot\pi)^2}{-2\theta_\sigma^2}}\right). \quad (12)$$

According to (4), $|\theta-\theta_\mu-\alpha\cdot\pi|\le(\pi/2)$ can be derived; thus, (12) can be reorganized

$$\sum_{x=0,x\neq\theta_{\mathrm{gt}}}^{\frac{M-1}{M}\cdot\pi}\operatorname{Exp}\left(L_{\mathrm{G}}(x)-L_{\mathrm{G}}\left(\theta_{\mathrm{gt}}\right)\right)\le\sum_{x=0,x\neq\theta_{\mathrm{gt}}}^{\frac{M-1}{M}\cdot\pi}\operatorname{Exp}\left(e^{\frac{\pi^2}{-8\theta_\sigma^2}}-e^{\frac{(\theta_{\mathrm{gt}}-\theta_\mu-\alpha\cdot\pi)^2}{-2\theta_\sigma^2}}\right). \quad (13)$$

In (13), $e^{\pi^2/(-8\theta_\sigma^2)}$ needs to be minimized; thus, θ_σ will converge to 0. In contrast, $e^{((\theta_{\mathrm{gt}}-\theta_\mu-\alpha\cdot\pi)^2)/(-2\theta_\sigma^2)}$ needs to be maximized. According to (4), this term can be rewritten as $e^{((\theta_{\mathrm{gt}}-\theta_\mu)^2)/(-2\theta_\sigma^2)}$, because α will equal to 0 when θ_μ is close to θ_{gt}. Therefore, θ_μ and θ_σ will converge to θ_{gt} and 1, respectively. Notably, as the deviation between the predicted θ_μ and θ_{gt} increases, θ_σ in $e^{((\theta_{\mathrm{gt}}-\theta_\mu)^2)/(-2\theta_\sigma^2)}$ will tend to increase, which is opposite to the direction of the optimization in $e^{\pi^2/(-8\theta_\sigma^2)}$. Therefore, the closer the θ_μ is to θ_{gt}, the lower the θ_σ will be, which demonstrates θ_σ can implicitly represent the quality of the predicted angle.

B. AQ-NMS

After obtaining the estimated quality of the predicted angle, i.e., $\mathrm{quality}_{\mathrm{angle}}$, we utilize it to build a comprehensive evaluation of detection results during the postprocessing, rectifying the misalignment between the classification score and the localization precision. In postprocessing, NMS is commonly implemented to eliminate redundant detection results, which employs classification score as the evaluation criterion for assessing a set of overlapping detection results, and those with low scores are suppressed. Since OBBs with low classification score may have high localization precision [38], [39], [40], classification score is inappropriate as the single keyword in the ranking. Considering angle plays a crucial role in the localization of OBBs, we, thus, propose an AQ-NMS, alleviating the aforementioned misalignment by introducing the estimated quality of angle.

Formally, for the arbitrary detection result, it can be represented as $\{x_{\mathrm{center}}, y_{\mathrm{center}}, w, h, \theta_\mu, \theta_\sigma, \mathrm{score}_{\mathrm{cls}}\}$. We here combine the classification score with the estimated quality of angle and then generate a composite evaluation score S as follows:

$$S=\xi\cdot\mathrm{score}_{\mathrm{cls}}+(1-\xi)\cdot\mathrm{quality}_{\mathrm{angle}} \quad (14)$$

where $\mathrm{score}_{\mathrm{cls}}$ is the classification score and ξ is the control factor. With AQ-NMS, the evaluation misalignment that leads to detection failure can be avoided. The detailed process is summarized in Algorithm 1.

Algorithm 1 AQ-NMS

Input: Initial detection results: $D=\{d_1,\ldots,d_N\}$; Classification Score: $C=\{c_1,\ldots,c_N\}$; Angle Quality: $Q=\{q_1,\ldots,q_N\}$; Trade-off factor: ξ; Thresholds: τ, ε.
Output: Detection results: O

```
O ← ∅
S ← ∅
// Combine classification score with angle quality
for each i in [1, ..., N] do
    if c_i > τ then
        s_i = ξ · c_i + (1 − ξ) · q_i
    else
        s_i = 0
    end if
    S ← S ∪ s_i
end for
// Perform standard Non-Maximum Suppression
O ← NMS(D, S, ε)
return O
```

C. Triple-A Loss

Objects with large aspect ratios, such as large vehicles and ships, are highly sensitive to the variations of angles, so slight deviations of angle will produce large differences in localization. In contrast, those with low aspect ratios, such as airplanes and storage tanks, are insensitive to the shifts of angle. However, the same supervision is often imposed on each object in the existing methods, which is unreasonable and incorrect. On the one hand, the localization precision of angle-insensitive objects is weakly correlated with angle; thus, it is redundant to impose excessive supervision on them. On the other hand, angle plays a significant role in localization precision for angle-sensitive objects; thus, more supervision should be applied. To this end, we propose a Triple-A Loss, for applying dynamic weights on different objects.

First, we refer to the influence degree of angle to localization precision as angle sensitivity and analyze the numerical relationship between it and object aspect ratio. We conduct a simulation of rotation variation for objects with different aspect ratios and record the varying process of IoU. Specifically, we set the short side length and angle of the initial OBB to 1 and 0, respectively, and rotate it from 0 to $\pi/2$ with a $\pi/180$ interval for recording the IoU variation. Afterward, we adjust its aspect ratio from 1 to 10 with a 0.2 interval and repeat the above process; thereby, a series of IoU values for OBB with different aspect ratios can be collected. As shown in Fig. 3, each colorful curve represents the IoU variation of the object under a kind aspect ratio, where the higher the curve, the lower the aspect ratio. It is obvious that the angle sensitivity significantly increases along with the aspect ratio raises. It can also be seen that for the object whose aspect ratio is close to 1, the variations in angle affect little on IoU, where the lowest IoU value satisfies the requirement of correct detection [44], which confirms our observation. Following the criterion reported in [44], we record the angle value of each object when its IoU is equal to 0.5. The correlation between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

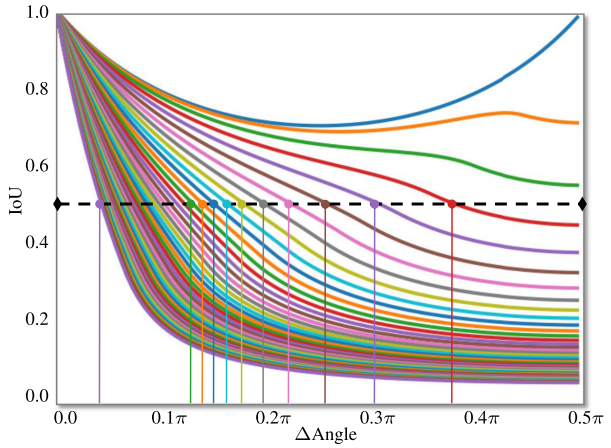


Fig. 3. Visualization of IoU variations of the object against different aspect ratios—the higher the curve is, the lower the aspect ratio holds.

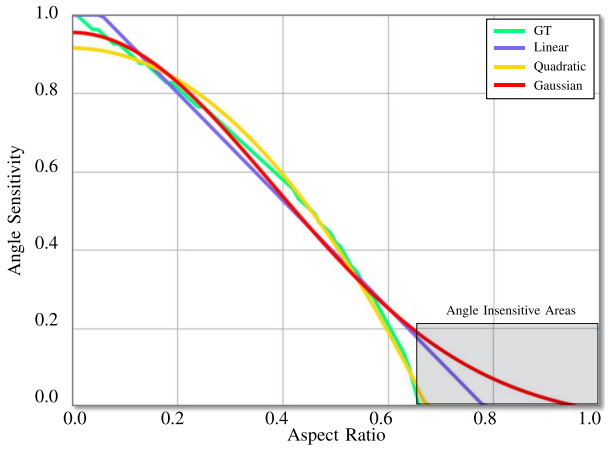


Fig. 4. Visualization of different angle sensitivity curves. When the aspect ratio is enough high, the IoU between the object and the ground truth is always greater than 0.5, regardless of how the object rotates. In angle-insensitive areas, we ensure that the values of the fitted function are always greater than 0.01.

the angle sensitivity and the ratio of objects is denoted by the green line in Fig. 4, where the ratio indicates the inverse of the aspect ratio.

Then, a series of functions is employed to approximate the tendency of the curve by applying the least squares method, including linear, quadratic, and Gaussian functions

$$\begin{cases} \mathrm{AS_L}(\mathrm{ratio}) = \lambda_{\mathrm{L}} \cdot \mathrm{ratio} + \zeta_{\mathrm{L}} \\ \mathrm{AS_Q}(\mathrm{ratio}) = \lambda_{\mathrm{Q}} \cdot \mathrm{ratio}^2 + \zeta_{\mathrm{Q}} \\ \mathrm{AS_G}(\mathrm{ratio}) = e^{-\lambda_{\mathrm{G}} \cdot \mathrm{ratio}^2} + \zeta_{\mathrm{G}} \end{cases} \tag{15}$$

where $\mathrm{AS_L}$, $\mathrm{AS_Q}$, and $\mathrm{AS_G}$ are the approximated tendency by using linear, quadratic, and Gaussian function, respectively. The parameters λ_{L}, λ_{Q}, λ_{G}, ζ_{L}, ζ_{Q}, and ζ_{G} are fit to -1.376, -2.013, -3.386, 1.078, 0.916, and -0.044 with the least squares method, respectively. In this way, we can adaptively adjust the trade-off of angle prediction for each object, reducing the redundant angle losses, as follows:

$$L_{\text{Aspect Ratio}} = -\mathrm{AS}_* \cdot \mathrm{CrossEntropy}\left(L_{\mathrm{G}}, L_{\mathrm{gt}}\right). \tag{16}$$

Finally, although AS_* suppresses the redundant losses, it does not differentiate between easy and hard instances, i.e., well-trained and insufficient-trained objects. Therefore, we take inspiration from [11] and further propose a Triple-A Loss to adaptively adjust the loss weights according to both the angle sensitivity and difficulty of objects, which is defined as follows:

$$\begin{aligned} L_{\text{TripleA}} &= \left(1 - p_{\theta_{\mathrm{gt}}}\right)^{\gamma*(1-\mathrm{AS}_*)} \cdot L_{\text{Aspect Ratio}} \\ &= -\mathrm{AS}_* \cdot \left(1 - p_{\theta_{\mathrm{gt}}}\right)^{\gamma*(1-\mathrm{AS}_*)} \mathrm{CrossEntropy}\left(L_{\mathrm{G}}, L_{\mathrm{gt}}\right) \end{aligned} \tag{17}$$

where $p_{\theta_{\mathrm{gt}}}$ is the probability of the predicted angle belonging to θ_{gt}. Instead of applying a constant parameter to control the penalty for difficult samples [11], we use angle sensitivity to replace it, which avoids the worthless attention on objects with low aspect ratios, alleviating the optimization misalignment.

D. Loss Function

The proposed method is a plug-and-play manner, and we here take Faster-RCNN [9] as an instance, which is optimized in an end-to-end way by using a multitask loss as follows:

$$L = \lambda_c \cdot L_{\mathrm{cls}} + \lambda_b \cdot L_{\mathrm{bbox}} + \lambda_\theta \cdot L_{\mathrm{TripleA}} \tag{18}$$

where L_{cls}, L_{bbox}, and L_{TripleA} represent the loss function of classification, bounding box regression, and angle prediction, respectively. The hyperparameter λ_c, λ_b, and λ_θ is used to balance the importance of terms. In our experiment, we set them to 2.0, 1.0, and 0.2, respectively. Since we do not perform additional operations on the classification branch and bounding box regression branch, we employ the loss functions of cross entropy and smooth $L1$ for them, respectively, keeping consistent with [9]. For angle prediction, we use Triple-A Loss presented in Section III-C.

IV. Datasets and Evaluation Metrics

In this section, we will introduce the datasets and evaluation metrics used in our experiments. To demonstrate the effectiveness of the proposed methodology, we evaluate our proposal on three public-oriented object detection datasets: DOTA [18], HRSC2016 [19], and ICDAR2015 [20].

A. DOTA

DOTA [18] is a large-scale dataset for object detection in RSIs. It contains 2806 images from different sensors and platforms. Each image is about 800 × 800 to 4000 × 4000 pixels in size and contains objects with a wide variety of scales, orientations, and shapes. These DOTA images are annotated by experts in aerial image interpretation using 15 common object categories, including plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter (HC), roundabout, soccer ball field, and basketball court. The fully annotated DOTA dataset contain 1 88 282 instances, which is annotated by an OBB. Half of the images are set as the training set, 1/6 as the validation set, and 1/3 as the test set. Similar to the existing methods, we split the original image in the dataset into 1024 × 1024 in size with an overlap of 100 pixels. After splitting, the training dataset includes 10 276 images, the validation dataset includes 2957 images, and the testing dataset includes 10 833 images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I

ABLATION STUDY OF AQE-DET IN TERMS OF mAP_{50}. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Model	AQE	Triple-A Loss	AQ-NMS	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP_{50}
AQE-Det	×	×	×	**90.01**	76.97	45.61	69.52	65.81	78.05	87.17	90.84	64.28	**87.95**	72.51	65.91	64.81	64.36	48.33	71.47
	✓	×	×	90.00	77.53	**47.54**	69.10	66.06	79.05	88.11	90.86	**69.56**	87.63	72.64	69.14	65.46	64.37	50.23	72.49
	✓	✓	×	90.00	78.85	47.18	72.31	66.10	79.45	88.22	**90.87**	69.00	87.68	77.84	70.01	65.23	64.81	**56.19**	73.58
	✓	✓	✓	89.55	**79.26**	47.30	**72.51**	**66.30**	**80.22**	**88.25**	90.85	68.88	87.68	**78.27**	**70.19**	**65.83**	**65.41**	55.45	**73.73**

TABLE II

ABLATION STUDY OF AQE-DET IN TERMS OF mAP_{75}. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Model	AQE	Triple-A Loss	AQ-NMS	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP_{75}
AQE-Det	×	×	×	66.23	27.72	6.29	39.76	26.15	38.52	34.72	79.92	49.33	55.30	46.32	**35.36**	12.39	5.30	5.88	35.28
	✓	×	×	69.61	31.16	7.94	41.48	26.80	44.74	42.63	81.02	52.90	54.75	**48.66**	34.37	15.40	11.60	10.47	38.24
	✓	✓	×	67.09	28.19	**11.77**	41.87	27.87	46.21	42.41	79.82	57.69	56.66	43.75	35.01	20.30	11.53	13.94	38.94
	✓	✓	✓	**69.87**	**33.21**	10.70	**42.03**	**28.14**	**46.26**	**42.76**	**83.98**	**57.76**	**56.73**	44.58	34.66	**20.96**	**11.92**	**15.93**	**39.97**

B. HRSC2016

HRSC2016 [19] is a challenging ship dataset that contains images from two scenarios: ships on the sea and ships near the shore. All the images are collected from six famous ports. In the dataset, the image resolution is between 2 and 0.4 m, the image size is between 300 × 300 and 1500 × 900, and most of them are larger than 1000 × 600. The training, validation, and test datasets consist of 436, 181, and 444 images, respectively.

C. ICDAR2015

ICDAR2015 [20] is the Challenge 4 of ICDAR 2015 Robust Reading Competition, which is commonly used for oriented scene text detection and spotting. In the dataset, the image size is 720 × 1280. The dataset contains 1500 images, including 1000 training images and 500 testing images.

D. Evaluation Metrics

In this article, the widely used measures in object detection, including precision (P), recall (R), $F1$ measure, and mean average precision (mAP), are adopted to evaluate the performance of the detector. Among them, mAP includes mAP_{50} and mAP_{75}. These two metrics indicate the mAP value under the IoU threshold of 0.5 and 0.75, respectively.

V. EXPERIMENTS

A. Implementation Details and Parameter Optimization

In this article, all the experiments are implemented under the PyTorch framework. For DOTA [18], HRSC2016 [19], and ICDAR2015 [20], the input image is resized to 1024 × 1024, 800 × 800, and 720 × 1280, respectively, and the pixels are padded with zeros to keep the aspect ratio. Then, the network is trained on NVIDIA GeForce GTX 1080Ti with 11-GB GPU memory, and the batch size is set to 4. For the network initialization, the weights pretrained on ImageNet are used to initialize the backbone (i.e., ResNet [49]), and other added layers are randomly initialized by a zero-mean normal distribution with a standard deviation of 0.01. SGD [50] is selected as the optimizer for our model, and the learning rate is set to 0.01 at the beginning, which is decreased by a factor of 10 at 8 and 11 epochs, respectively, and the model is trained with 24 epochs in total. The training steps per epoch for DOTA [18], HRSC2016 [19], and ICDAR2015 [20] are 27k, 4k, and 10k, respectively. In the training phase, we only use simple data augmentations, including random horizontal flipping, random rotation, and random scale, to enhance the robustness of the detector. In the inference phase, the output threshold is set to 0.001.

B. Ablation Study

We perform the ablation study on the DOTA and HRSC2016 datasets to verify the effectiveness of each proposed component. Faster-RCNN [9] is selected as the baseline, and the same training strategies are adopted in all the experiments. The improved Faster-RCNN by applying our proposed method is called AQE-Det.

1) Effectiveness of AQE: To validate the superiority of AQE, we conduct experiments on the DOTA dataset with a complex variety of objects, as well as on the HRSC2016 dataset with lots of high aspect-ratio objects.

For DOTA, as shown in Tables I and II, in comparison with the baseline, adding AQE yields a salient improvement of 1.02%/2.96% on mAP_{50}/mAP_{75}. This indicates that our AQE can make better predictions for multicategory objects, even if the scenario is complex. Meanwhile, the experimental results exhibit a larger improvement in terms of mAP_{75}. Our proposed approach achieves significant improvement in dealing with the objects sensitive to angles, i.e., large vehicle, ship, and harbor, which outperforms the baseline by 6.22%, 7.91%, and 3.01%, respectively. This confirms that our AQE is much beneficial to angle-sensitive objects.

For HRSC2016, as shown in Table III, inserting AQE results in a significant improvement compared with the benchmark Faster-RCNN, which brings 8.31% improvements on mAP_{50}. This indicates that a large performance gain is acquired by using AQE, which addresses the representation misalignment. Furthermore, the experimental results observe 11.20% improvements in terms of the more precise metric mAP_{75}, which confirms the proposed angle prediction approach greatly enhances the localization precision. It is worth noting that the improvement increases with the raise of IoU thresholds. To analyze how much computational costs are introduced by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ABLATION STUDY OF AQE-DET. ALL MODELS ARE EVALUATED ON HRSC2016 TEST SET. AUG. DENOTES USING DATA AUGMENTATION

Backbone	AQE	Triple-A Loss	AQ-NMS	Aug.	mAP_{50}	mAP_{75}
ResNet-50	×	×	×	×	78.24	38.92
	✓	×	×	×	86.55	50.12
	✓	✓	×	×	87.32	54.46
	✓	✓	✓	×	88.14$_{\uparrow 9.90}$	58.47$_{\uparrow 19.55}$
	×	×	×	✓	88.31	50.04
	✓	✓	✓	✓	**89.76**$_{\uparrow 1.45}$	**75.12**$_{\uparrow 25.08}$
ResNet-101	✓	✓	✓	✓	**90.02**	**76.76**

TABLE IV

MODEL PARAMETERS AND TRAINING COSTS OF AQE-DET. ALL MODELS ARE EVALUATED ON HRSC2016 TEST SET. REG. AND CLS. DENOTE REGRESSION- AND CLASSIFICATION-BASED METHODS, RESPECTIVELY

Method	Params	GFLOPs	Training Time	mAP_{50}	mAP_{75}
Reg. [11]	36.13 M	128.09	0.2495s/iter	83.21	50.72
Cls. [33]	40.15 M	181.79	0.3870s/iter	85.40	60.20
AQE-Det	36.15 M	128.37	0.3468s/iter	86.28	72.32

the proposed AQE, we construct the experimental comparisons in Table IV, wherein RetinaNet [11] is selected as the baseline, i.e., regression-based method, and the input image is resized to 800 × 800. Obviously, our method can greatly improve the detection performance with almost no extra parameters (36.15 versus 36.13 M), which demonstrates its effectiveness.

2) Effectiveness of Triple-A Loss: To show the effect of Triple-A Loss, we conduct experiments on DOTA and HRSC2016.

For DOTA, using AQE, we observe 1.09% and 0.70% improvements on mAP_{50} and mAP_{75}, respectively, as shown in Tables I and II. It can be seen that by reducing the redundant supervision of angle prediction, the optimizer is guided in a more meaningful direction. Therefore, the rare object, such as HC, can be better recognized, whose mAP_{50} is significantly increased by 5.96%. As a trade-off, Triple-A Loss slightly hurts high-precision performance for those objects with low aspect ratios, which is attributed to the unavoidable reduction of attention. On the whole, an overall improvement is achieved by adaptively controlling the weights of samples.

For HRSC2016, based on the baseline with AQE, the results in Table III demonstrate that utilizing Triple-A Loss brings 0.77% and 4.34% improvements on mAP_{50} and mAP_{75}, respectively. This is due to the fact that it is inappropriate to apply the same supervision on ships with large differences in aspect ratio; meanwhile, the weights of the difficult training samples are enhanced, strengthening the detector's robustness.

3) Effectiveness of AQ-NMS: After adding both AQE and Triple-A Loss, the quality of the estimated angles is utilized to rerank the detection results, which further alleviates the misalignment between classification scores and localization precision.

For DOTA, as shown in Tables I and II, the improvements demonstrate the effectiveness of AQ-NMS. Especially, in terms of mAP_{75}, utilizing AQ-NMS brings 1.03% improvement. In comparison with the baseline, almost all the indicators are improved. For HRSC2016, the experimental results show

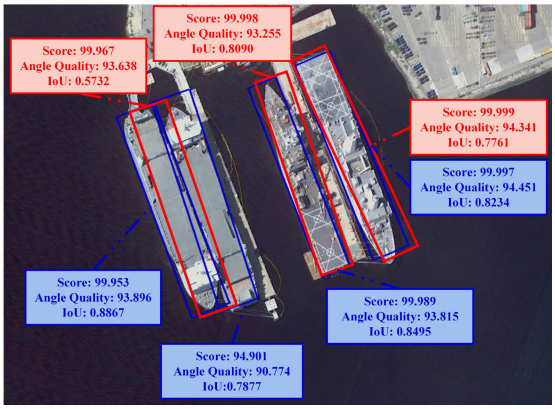


Fig. 5. Comparison of AQ-NMS Results. The detection results with and without AQ-NMS are denoted by blue and red rectangles.

0.92% and 4.01% gains on mAP_{50} and mAP_{75}, respectively. To further illustrate the effectiveness of AQ-NMS, we visualize the comparison of the detection results in Fig. 5. As we can see, the red OBBs are with high classification scores, but low localization accuracy suppresses the more precise blue OBBs when solely relying on the classification score, while AQ-NMS addresses this issue.

Integrating all the above modules significantly outperforms the baseline. In detail, without any data augmentation, our AQE-Det achieves a 19.55% improvement in terms of mAP_{75} on HRSC2016. To further ensure the effectiveness of the proposed method, under the same data augmentation, such as random flip, random rotate, and multiscale training, AQE-Det gains a 25.08% improvement on mAP_{75}. This confirms that even based on a powerful baseline model, AQE-Det can further improve its performance. Moreover, we visualize the precision–recall curves of AQE-Det, as shown in Fig. 6. The visualization results demonstrate that each module has significantly improved the baseline. For the DOTA dataset, AQE-Det outperforms the baseline by 2.26% and 4.89% in terms of mAP_{50} and mAP_{75}, respectively. This indicates that the proposed method is effective for arbitrary-oriented objects. For comparison, we visualize the detection results of the baselines and AQE-Det, which shows using AQE-Det can bring distinct advantages in angle prediction, as shown in Figs. 7 and 8.

C. Results on DOTA

We compare our AQE-Det with the common methods on the DOTA dataset [18], including [12], [13], [14], [16], [18], [26], [52], [56], [57]. Furthermore, the state-of-the-art methods are also selected to make comparison [17], [33], and [47].

As shown in Table V, by using ResNet-50 as the backbone, AQE-Det achieves 78.19% mAP_{50}, which significantly outperforms most of the other methods. When employing the backbone ResNet-101, our AQE-Det improves mAP_{50} by 0.16% and rises up to 78.35%. In comparison with the official baseline [18], our proposed method achieves a remarkable improvement by 25.42% in mAP_{50}, which confirms the superiority of our model. Using the RoI Transformer [15] as the base detector, our AQE-Det can achieve 80.87% mAP_{50}, which outperforms all the existing methods, confirming its superiority.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

TABLE V

OVERALL PERFORMANCE COMPARISONS ON DOTA TEST SET. † MEANS THE RESULTS FROM MMROTATE [45]

Model	Backbone	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP_{50}
FR-O [18]	ResNet-101	79.09	69.12	17.17	63.49	34.20	37.16	36.20	89.19	69.60	58.96	49.4	52.52	46.69	44.80	46.30	52.93
R-DFPN [14]	ResNet-101	80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88	57.94
R^2CNN [12]	ResNet-101	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [13]	ResNet-101	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
ICN [26]	ResNet-101	81.40	74.30	47.70	70.30	64.90	67.80	70.00	90.80	79.10	78.20	53.60	62.90	67.00	64.20	50.20	68.20
RoI-Transformer [15]	ResNet-101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [45]	ResNet-101	87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
DRN [46]	Hourglass-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
DAL [36]	ResNet-101	88.61	79.69	46.27	70.37	65.89	76.10	78.53	90.84	79.98	78.41	58.71	62.02	69.23	71.32	60.65	71.78
SCRDet [31]	ResNet-101	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
R^3Det [16]	ResNet-152	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
Gliding Vertex [17]	ResNet-101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
Mask OBB [47]	ResNeXt-101	89.56	**85.95**	54.21	72.90	76.52	74.16	85.63	89.85	83.81	86.48	54.89	69.64	73.94	69.06	63.32	75.33
CSL [33]	ResNet-152	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	76.17
DCL [34]	ResNet-101	89.14	83.93	53.05	72.55	78.13	81.97	86.94	90.36	85.98	86.94	66.19	65.66	73.72	71.53	68.69	76.97
ReDet [48]	ReResNet-50	88.81	82.48	**60.83**	**80.82**	78.34	**86.06**	88.31	90.87	**88.77**	87.03	68.65	66.90	79.26	79.71	**74.67**	80.10
AQE-Det	ResNet-50	89.35	84.21	55.70	76.76	74.05	78.78	87.45	90.84	88.05	86.10	67.60	67.17	75.76	77.22	73.77	78.19
	ResNet-101	89.49	85.89	55.46	77.32	74.17	80.33	87.65	90.82	86.99	86.52	66.58	69.06	77.12	78.35	69.51	78.35
RoI-Transformer†	ResNet-50	88.76	84.44	57.53	78.65	79.67	85.49	88.26	**90.90**	87.05	**88.20**	69.70	68.77	78.89	81.48	70.07	79.86
RoI-Transformer†+AQE-Det	ResNet-50	89.23	85.35	60.26	80.13	**80.27**	85.58	**88.46**	90.88	87.51	88.07	**69.80**	70.32	**80.97**	**82.16**	74.03	**80.87**

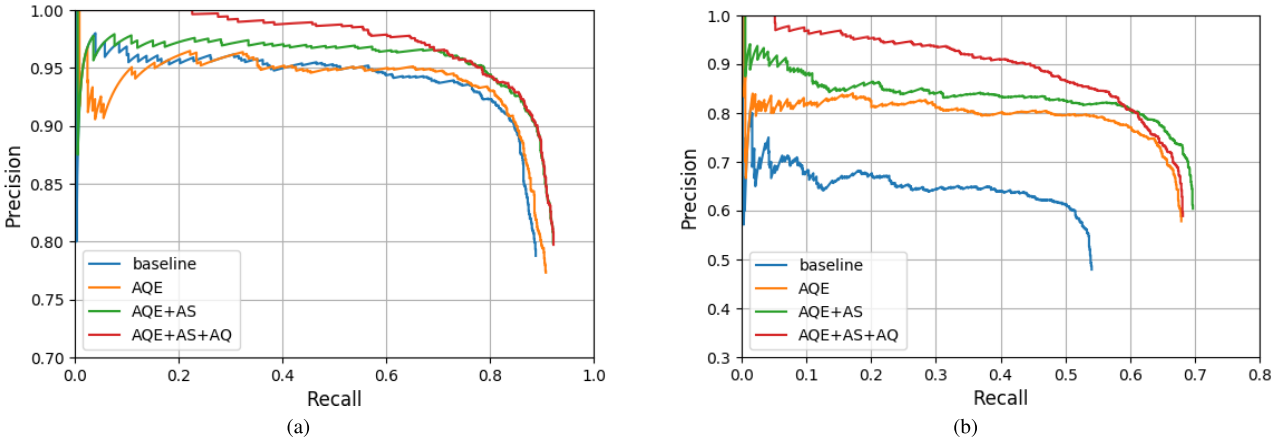


Fig. 6. Precision–recall curve of AQE-Det on HRSC2016 in terms of (a) mAP_{50} and (b) mAP_{75}. AS and AQ indicate Triple-A Loss and AQ-NMS, respectively.

Moreover, compared with the state-of-the-art methods [17], [33], [47], AQE-Det outperforms them with 3.33%, 3.02%, and 2.18% improvements on mAP_{50}, respectively, which validate the advantages of the proposed method.

For objects with large aspect ratios, such as large vehicle, ship, and harbor, AQE-Det achieves the state-of-the-art performance of 80.33%, 87.65%, and 77.12%, respectively. In addition, for the rare objects, i.e., swimming pool and HC, benefiting from Triple-A Loss, it achieves the best performance of 78.35% and 73.77%, respectively. It is worth noting that our AQE-Det is a plug-and-play technology, which can be easily combined with arbitrary detectors. As shown in Table VI, inserting the AQE module can bring comprehensive improvements of mAP_{50}/mAP_{75} on the existing methods [9], [11], [15], [48]. Some visualization results of AQE-Det on DOTA are shown in Fig. 9.

D. Results on HRSC2016

We further evaluate our model on the HRSC2016 dataset, which includes numerous ships with large aspect ratios.

TABLE VI

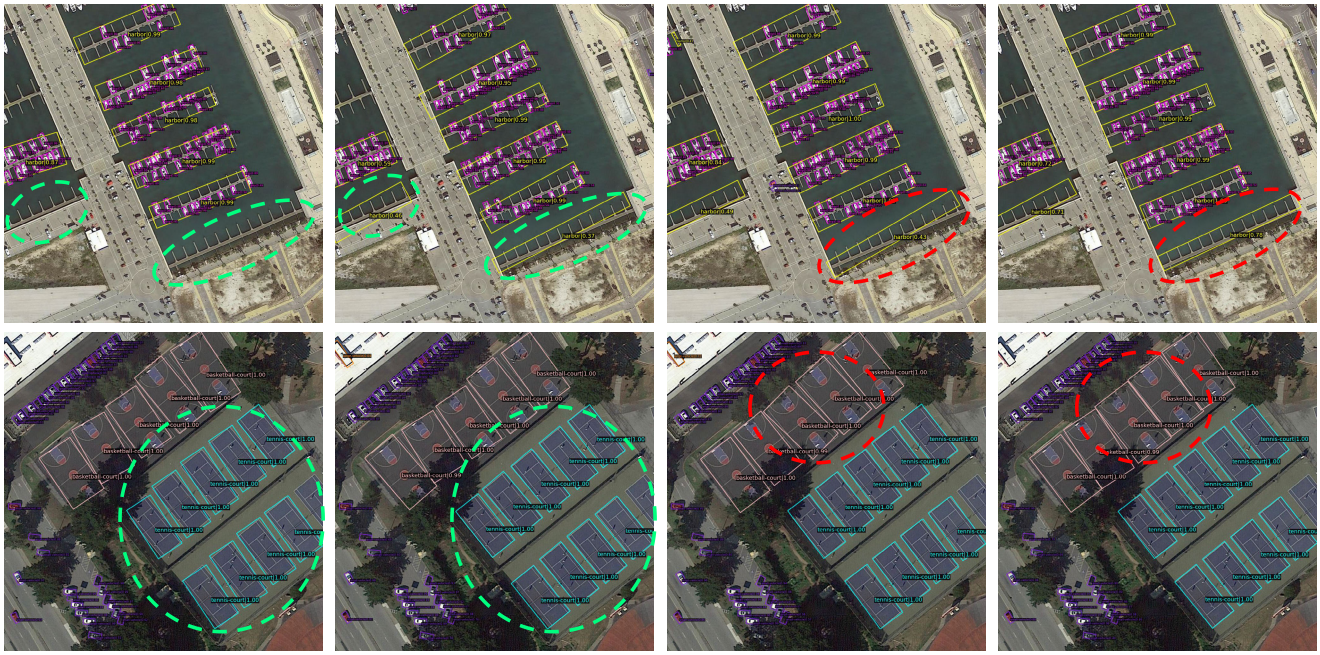
PORTABILITY ANALYSIS OF AQE-DET. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Method	Backbone	AQE	mAP_{50}	mAP_{75}
RetinaNet [11]	ResNet-50	×	64.31	36.40
		✓	66.86$_{\uparrow 2.55}$	38.04$_{\uparrow 1.64}$
Faster-RCNN [9]		×	71.47	35.28
		✓	73.73$_{\uparrow 2.26}$	39.97$_{\uparrow 4.69}$
RoI-Transformer [15]		×	75.54	47.02
		✓	76.31$_{\uparrow 0.77}$	48.33$_{\uparrow 1.31}$
ReDet [48]	ReResNet-50	×	76.52	51.73
		✓	76.81$_{\uparrow 0.29}$	52.03$_{\uparrow 0.30}$

To demonstrate the effectiveness of the proposed method, we conduct the comparison experiments with other state-of-the-art methods [15], [16], [17], [30], [33], [34], [36].

The overall comparison performance is reported in Table VII. As we can see, the proposed AQE-Det with ResNet-50 as backbone has already achieved an mAP_{50}

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.



Faster-RCNN [9] | Faster-RCNN [9]+AQE | RoI-Transformer [15] | RoI-Transformer [15]+AQE

Fig. 7. Visualization comparison of detection results in DOTA. Obviously, our proposed method can significantly ameliorate the existing oriented object detectors, such as Faster-RCNN [9] and RoI Transformer [15].

TABLE VII
OVERALL PERFORMANCE COMPARISONS ON HRSC2016 TEST SET

Model	Backbone	Image Size	mAP_{50}
Fast-RCNN+SRBBS [51]	VGG-16	-	55.70
BL2 [51]	VGG-16	-	69.60
R^2CNN [12]	ResNet-101	800×800	73.07
IENet [52]	ResNet-101	1024×1024	75.01
RC1&RC2 [51]	VGG-16	-	75.70
RRPN [13]	ResNet-101	800×800	79.05
RRD [27]	VGG-16	384×384	84.30
RoI-Transformer [15]	ResNet-101	512×800	86.20
Gliding Vertex [17]	ResNet-101	-	88.20
R^3Det [16]	ResNet-101	800×800	89.26
DCL [34]	ResNet-101	800×800	89.46
GRS-Det [30]	ResNet-101	800×800	89.57
CSL [33]	ResNet-101	800×800	89.62
DAL [36]	ResNet-101	800×800	89.77
AQE-Det	ResNet-50	800×800	89.76
	ResNet-101	800×800	**90.02**

TABLE VIII
OVERALL PERFORMANCE COMPARISONS ON ICDAR2015 TEST SET

Model	Precision	Recall	F1-Measure
CTPN [53]	74.22	51.56	60.85
Seglink [54]	73.10	76.80	75.00
RRPN [13]	82.17	73.23	77.44
EAST [55]	78.33	83.27	80.72
SCRDet [31]	81.30	78.90	80.08
RRD [27]	85.60	79.00	82.20
DAL [36]	84.40	80.50	82.40
CSL [33]	84.30	83.00	83.65
Baseline	82.12	83.12	82.62
AQE-Det	**85.65**	**83.53**	**84.58**

of 89.76%, which outperforms the official baseline of HRSC2016 (BL2 [51]) by 34.06%. This confirms the significant superiority of AQE-Det. Furthermore, in comparison with the state-of-the-art methods [15], [16], [17], [30], [33], our AQE-Det outperforms them by 3.56%, 1.56%, 0.5%, 0.19%, and 0.14%, respectively. Moreover, using the ResNet-101 [49] as the backbone, our AQE-Det further improves mAP_{50} by 0.26% and rises up to 90.02%, which achieves the state-of-the-art performance. To further show the performance of AQE-Det, some detection results are visualized in Fig. 10. Most of the ships own large aspect ratios, which are sensitive to variations of angles. It can be seen that our AQE-Det can precisely detect them, overcoming the representation and evaluation misalignments.

E. Results on ICDAR2015

To verify the generalization ability of AQE-Det, we also evaluate our model on the text detection benchmark ICDAR2015, which includes numerous oriented text and complex background. To demonstrate the performance of the proposed method, we conduct the comparison experiments with state-of-the-art methods [31], [33], [36] and several common methods [13], [27], [53], [54], [55].

The overall comparison performance is reported in Table VIII. It can be seen that the proposed AQE-Det achieves an $F1$ measure of 84.58%, which outperforms the state-of-the-art methods [31], [33], [36] by 4.50%, 2.18%, and 0.93%,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.



Fig. 8. Visualization comparison of detection results in HRSC2016. Obviously, Faster-RCNN [9] and RoI Transformer [15] are inferior to our AQE-Det, which avoids the misalignment issues.



Fig. 9. Some qualitative visualization results on the DOTA dataset.

respectively. Meanwhile, in comparison with our baseline, AQE-Det improves $F1$ measure by 1.96%, especially a 3.53% improvement in terms of precision. This confirms the excellent generalization ability of AQE-Det.

F. Parameter Analysis

In this part, we conduct three experiments to analyze the parameters in AQE, Triple-A Loss, and AQ-NMS, respectively. The remaining variable parameters take optimal values in all the experiments. The experiments are implemented on the DOTA validation set.

1) Evaluation of Angle Resolution: To analyze the relationship between angle resolution and detection performance, we perform comparison experiments with different angle resolutions. In this experiment, the angle resolutions R_θ is set to 10, 5, and 1 for comparison. Table IX shows the performance with each angle resolution.

It can be seen that when the angle resolution is 10, the training time of a single batch is only 0.3440 s; however, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

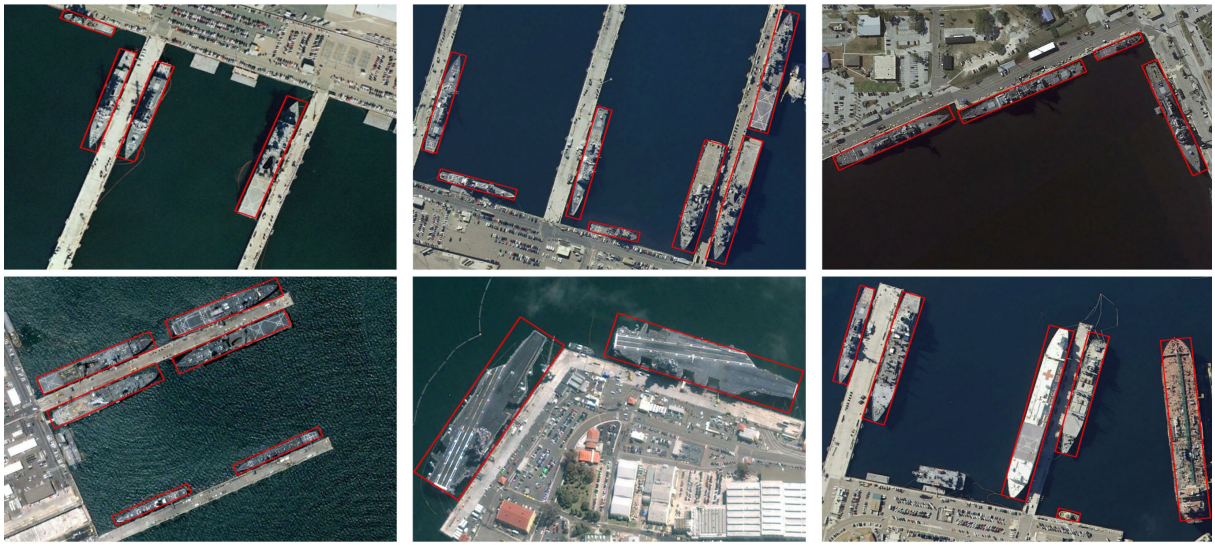


Fig. 10. Visualization of detection results from AQE-Det in HRSC2016.

TABLE IX
PARAMETER ANALYSIS ON AQE. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Method	R_θ	Training Time/batch	mAP_{50}	mAP_{75}
AQE-Det	10	**0.3440s**	72.29	33.04
	5	0.3442s	72.64	37.50
	1	0.3468s	**73.73**	**39.97**

TABLE X
PARAMETER ANALYSIS ON TRIPLE-A LOSS. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Model	Angle Sensitivity	γ	mAP_{50}	mAP_{75}
AQE-Det	Linear Function	5	72.75	38.55
	Quadratic Function	5	73.12	37.98
	Gaussian Function	5	**73.73**	**39.97**
	Gaussian Function	2	73.20	38.78
	Gaussian Function	10	73.25	39.16

TABLE XI
PARAMETER ANALYSIS ON AQ-NMS. ALL MODELS ARE EVALUATED ON THE DOTA VALIDATION SET

Method	AQ-NMS	ξ	mAP_{50}	mAP_{75}
AQE-Det	×	0	73.58	38.94
	✓	0.05	73.69	39.60
	✓	0.1	**73.73**	**39.97**
	✓	0.2	73.58	39.74
	✓	0.5	72.63	39.58

lowest mAP is acquired. This is because low-angle resolution leads to inaccurate prediction, especially for large aspect ratio objects. When the angle resolution is 5, mAP_{50} and mAP_{75} raise up to 72.64% and 37.50%, respectively. Since our AQE only needs a little computation cost during the training phase, the training time almost unchanged. When the angle resolution equals 1, it acquires the highest performance and keeps the same training time. This confirms that our AQE significantly improves the performance with almost no computational costs.

2) Evaluation of Triple-A Loss: The angle sensitivity and γ are the keys to Triple-A Loss, so the setting of them is essential to performance. Therefore, we carry out five experiments to select better parameters.

As described in Section III-C, three functions are chosen to represent the angle sensitivity. The results in Table X show that the Gaussian function outperforms the others in terms of overall performance, which indicates the fitted Gaussian function can better characterize angle sensitivity along with aspect ratios. For γ, we select 2, 5, and 10 to conduct comparison experiments. As shown in Table X, when the γ is set to 5, we can achieve the best performance of 73.73%. The results verify the gamma 5 exhibits better trade-off between simple and difficult instances.

3) Evaluation of AQ-NMS: ξ is the crucial parameter in AQ-NMS. To choose a better trade-off, we conduct comparison experiments on the DOTA dataset.

As shown in Table XI, we explore the performance of the different settings, where ξ is set to 0, 0.05, 0.1, 0.2, and 0.5, respectively. It can be seen that when ξ takes 0.1, our AQE-Det achieves the highest performance. With the help of AQ-NMS, 0.15% improvement of mAP_{50} is brought; more especially, mAP_{75} obtains a 0.66% improvement. In total, along with ξ, the performance first increases and then decreases, while mAP_{75} is consistently much higher than the case without AQ-NMS, confirming the effectiveness of AQ-NMS.

VI. CONCLUSION

In this article, we proposed a novel angle prediction approach, called AQE. Considering a series of misalignments in the existing angle prediction methods, our AQE transforms the angle prediction problem into a distribution estimation task to eliminate the representation misalignment and implicitly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

estimate the quality of the predicted angle, which greatly improves the precision of angle prediction with almost no additional computation. Building upon the estimated AQ, an AQ-NMS is designed to rectify the evaluation misalignment by comprehensively assessing the detection results. Next, a Triple-A Loss is proposed to assist in the optimization of the angle prediction task. Finally, experiments on the challenging remote sensing-oriented detection datasets DOTA and HRSC2016 and the text detection benchmark ICDAR2015 are performed to demonstrate the effectiveness of the proposed method.

In the future, we will generalize our approach to other applications in RSIs, such as change detection, and object tracking.

REFERENCES

[1] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[2] T. Zhang et al., "Foreground refinement network for rotated object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610013.

[3] W. Ma et al., "Feature split–merge–enhancement network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616217.

[4] F. Sun, T. Kong, W. Huang, C. Tan, B. Fang, and H. Liu, "Feature pyramid reconfiguration with consistent loss for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5041–5051, Oct. 2019.

[5] Y. Li and S. Wang, "HAR-Net: Joint learning of hybrid attention for single-stage object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 3092–3103, 2020.

[6] G. Wang, X. Zhang, Z. Peng, X. Tang, H. Zhou, and L. Jiao, "Absolute wrong makes better: Boosting weakly supervised object detection via negative deterministic information," in *Proc. 31st Int. Joint Conf. Artif. Intell.*, Jul. 2022, pp. 1378–1384.

[7] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[8] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[10] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[11] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[12] Y. Jiang et al., "R2CNN: Rotational region CNN for orientation robust scene text detection," 2017, *arXiv:1706.09579*.

[13] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[14] X. Yang et al., "Automatic ship detection in remote sensing images from Google Earth of complex scenes based on multiscale rotation dense feature pyramid networks," *Remote Sens.*, vol. 10, no. 1, p. 132, Jan. 2018.

[15] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for detecting oriented objects in aerial images," 2018, *arXiv:1812.00155*.

[16] X. Yang, Q. Liu, J. Yan, and A. Li, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[17] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[18] G. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[19] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 2. Beijing, China: Institute of Automation Chinese Academy of Sciences, 2017, pp. 324–331.

[20] D. Karatzas et al., "ICDAR 2015 competition on robust reading," in *Proc. 13th Int. Conf. Document Anal. Recognit. (ICDAR)*, Aug. 2015, pp. 1156–1160.

[21] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[22] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands: Springer, 2016, pp. 21–37.

[23] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[24] Y. Zhu, C. Zhao, H. Guo, J. Wang, X. Zhao, and H. Lu, "Attention CoupleNet: Fully convolutional attention coupling network for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 113–126, Jan. 2019.

[25] T. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[26] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. Asian Conf. Comput. Vis.* Perth, WA, Australia: Springer, 2018, pp. 150–165.

[27] M. Liao, Z. Zhu, B. Shi, G. Xia, and X. Bai, "Rotation-sensitive regression for oriented scene text detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5909–5918.

[28] Z. Huang, W. Li, X.-G. Xia, and R. Tao, "A general Gaussian heatmap label assignment for arbitrary-oriented object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 1895–1910, 2022.

[29] L. Hou, K. Lu, and J. Xue, "Refined one-stage oriented object detection method for remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 1545–1558, 2022.

[30] X. Zhang, G. Wang, P. Zhu, T. Zhang, C. Li, and L. Jiao, "GRS-Det: An anchor-free rotation ship detector based on Gaussian-mask in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3518–3531, Apr. 2021.

[31] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8232–8241.

[32] W. Qian, X. Yang, S. Peng, Y. Guo, and J. Yan, "Learning modulated loss for rotated object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2458–2466.

[33] X. Yang and J. Yan, "Arbitrary-oriented object detection with circular smooth label," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 677–694.

[34] X. Yang, L. Hou, Y. Zhou, W. Wang, and J. Yan, "Dense label encoding for boundary discontinuity free rotation detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15819–15829.

[35] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11830–11841.

[36] Q. Ming, Z. Zhou, L. Miao, H. Zhang, and L. Li, "Dynamic anchor learning for arbitrary-oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2355–2363.

[37] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 923–932.

[38] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, "Acquisition of localization confidence for accurate object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 784–799.

[39] Y. He, C. Zhu, J. Wang, M. Savvides, and X. Zhang, "Bounding box regression with uncertainty for accurate object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2888–2897.

[40] J. Choi, D. Chun, H. Kim, and H. Lee, "Gaussian YOLOv3: An accurate and fast object detector using localization uncertainty for autonomous driving," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 502–511.

[41] Y. Lee et al., "Localization uncertainty estimation for anchor-free object detection," 2020, *arXiv:2006.15607*.

[42] S. Wu, X. Li, and X. Wang, "IoU-aware single-stage object detector for accurate localization," *Image Vis. Comput.*, vol. 97, May 2020, Art. no. 103911.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.

[43] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. NeurIPS*, 2020, pp. 21002–21012.
[44] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
[45] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019.
[46] X. Pan et al., "Dynamic refinement network for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11207–11216.
[47] J. Wang, J. Ding, H. Guo, W. Cheng, T. Pan, and W. Yang, "Mask OBB: A semantic attention-based mask oriented bounding box representation for multi-category object detection in aerial images," *Remote Sens.*, vol. 11, no. 24, p. 2930, Dec. 2019.
[48] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.
[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[50] N. Qian, "On the momentum term in gradient descent learning algorithms," *Neural Netw.*, vol. 12, no. 1, pp. 145–151, Jan. 1999.
[51] Z. Liu, J. Hu, L. Weng, and Y. Yang, "Rotated region based CNN for ship detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 900–904.
[52] Y. Lin, P. Feng, J. Guan, W. Wang, and J. Chambers, "IENet: Interacting embranchment one stage anchor free detector for orientation aerial object detection," 2019, *arXiv:1912.00969*.
[53] Z. Tian, W. Huang, T. He, P. He, and Y. Qiao, "Detecting text in natural image with connectionist text proposal network," in *Proc. Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands: Springer, 2016, pp. 56–72.
[54] B. Shi, X. Bai, and S. Belongie, "Detecting oriented text in natural images by linking segments," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2550–2558.
[55] X. Zhou et al., "EAST: An efficient and accurate scene text detector," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5551–5560.
[56] H. Wei, Y. Zhang, Z. Chang, H. Li, H. Wang, and X. Sun, "Oriented objects as pairs of middle lines," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 268–279, Nov. 2020.
[57] Y. Zhu, J. Du, and X. Wu, "Adaptive period embedding for representing oriented objects in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7247–7257, Oct. 2020.



Guanchun Wang received the B.S. degree in intelligent science and technology from Xidian University, Xi'an, China, in 2019, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education.

His research interests include deep learning, weakly supervised learning, object detection, and remote sensing image analysis.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in computer application technology from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree in pattern recognition and intelligent system from the School of Electronic Engineering, Xidian University, in 2006.

From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA. She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Peng Zhu received the B.S. degree from Xidian University, Xi'an, China, in 2017, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision and remote sensing image analysis.



Xu Tang (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from Xidian University, Xi'an, China, in 2007, 2010, and 2017, respectively, all in electronic circuit and system.

From 2015 to 2016 he was a Joint Ph.D. Student along with Prof. W. J. Emery at the University of Colorado at Boulder, Boulder, CO, USA. He is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include remote sensing image content-based retrieval and reranking, hyperspectral image processing, remote sensing scene classification, and object detection.



Puhua Chen (Senior Member, IEEE) received the B.S. degree in environmental engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2009, and the Ph.D. degree in circuit and system from Xidian University, Xi'an, China, in 2016.

She is currently a Lecturer with the School of Artificial Intelligence, Xidian University. Her research interests include machine learning, pattern recognition, and remote sensing image interpretation.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Artificial Intelligence, Xidian University, Xi'an, where he is the Director of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education. He is in charge of about 40 important scientific research projects and has authored over 20 monographs and 100 papers in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.



Huiyu Zhou received the B.Eng. degree in radio technology from the Huazhong University of Science and Technology, Wuhan, China, in 1990, the M.Sc. degree in biomedical engineering from the University of Dundee, Dundee, U.K., in 2002, and the Ph.D. degree in computer vision from Heriot-Watt University, Edinburgh, U.K., in 2006.

He is currently a Full Professor with the School of Computing and Mathematical Sciences, University of Leicester, Leicester, U.K. He has authored or coauthored over 380 peer-reviewed papers in the field.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:20:09 UTC from IEEE Xplore. Restrictions apply.